Abhishek Kumar

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Education

Vellore Institute of Technology, Vellore **2023 – 2027**
*Bachelor of Technology in Information Technology, CGPA: **8.16*** *Vellore, Tamil Nadu*

Experience

Software Development Intern **May 2026 – Jun 2026**
Ortect Organics

- Built **6+ responsive pages and reusable components** for the Ortect Organics company website using the **MERN stack** (MongoDB, Express.js, React.js, Node.js), working in 2-week Agile sprints.
- Integrated **5+ REST API endpoints** between the React frontend and Express.js backend to load and update website data dynamically, replacing manual content updates.
- Improved the Lighthouse accessibility score **from about 75 to 90+** by fixing 5+ issues such as missing alt text, low color contrast, and unlabeled form fields, and fixed layout issues across mobile, tablet, and desktop screens.

Projects

AgentFlow AI – Multi-Agent Market Intelligence Platform | Live Demo GitHub | **May 2026**
Python, FastAPI, LangGraph, Google Gemini, ChromaDB, PostgreSQL, React, TypeScript, Docker

- Built a **6-node LangGraph multi-agent pipeline** with FastAPI and Google Gemini that collects market data from Amazon, YouTube, and Reddit, runs sentiment, keyword, and topic analysis, and **auto-generates PDF/DOCX reports**.
- Implemented a **RAG pipeline** using Gemini embeddings stored in ChromaDB to answer follow-up questions on each report **with source citations**, returning a fallback reply instead of an unsupported answer.
- Developed a FastAPI + PostgreSQL backend with **29 REST endpoints**, JWT authentication, role-based access control, and prompt-injection checks, covered by **190 automated tests** running on GitHub Actions CI with Docker Compose.

CommerceLens – E-Commerce Revenue & Retention Analytics | Live Demo GitHub | **Sep 2026**
Python, SQL, DuckDB, Power BI, DAX, LightGBM, SHAP, Streamlit

- Analyzed **99,441 orders** across 9 tables of a Brazilian e-commerce marketplace using **DuckDB SQL (joins, aggregations, CTEs, window functions)**, and built a **Power BI dashboard** with **DAX measures** to track revenue, delivery and review KPIs for stakeholders.
- Ran a **scenario analysis** on delivery delays: late orders received 1–2 star reviews **62% of the time vs. 9%** for on-time orders; a what-if slicer estimates **∼1,700 fewer low reviews** if late deliveries were halved.
- Trained a **LightGBM** model to flag orders at risk of a 1–2 star review (**ROC-AUC 0.67**), explained the drivers with SHAP, and deployed it as a Streamlit app.

NSE Alpha Signal Research Framework GitHub | **Oct 2026**
Python, pandas, NumPy, statsmodels, scikit-learn, Plotly

- Designed a **cross-sectional alpha research pipeline** scoring **43 NSE stocks** daily on 4 signals (pairs mean-reversion, momentum, volume surprise, reversal), evaluated by rank IC, decile spread, turnover, and Newey–West t-statistics.
- Applied a formation / validation / held-out test split with a no-lookahead test; the pairs signal showed a significant IC (***t*-stat +2.12**), yet all 4 signals had a **negative Sharpe** after 10 bps/side costs.
- Backtested **long/short decile portfolios** with a 1-day execution lag, documented assumptions such as survivorship bias, and compared signals in an interactive Plotly **dashboard report**.

Achievements

- **Semi-Finalist, HackBattle Hackathon** (200+ teams) – IEEE CS Chapter, VIT; built and pitched a prototype **Sep 2024**

Leadership & Extracurricular

Technology and Gaming Club, VIT **Aug 2024 – Apr 2025**
Senior Core – Point of Contact (POC) *Vellore, Tamil Nadu*

- Managed club events as the Point of Contact and led TAG Clash and OverClock with a **40-member team**, handling budgeting, sponsorships, stakeholder coordination, and tournament scheduling.

Technical Skills

Languages: Python, SQL, Java, C++, JavaScript, TypeScript
Data Analysis & Visualization: pandas, NumPy, DuckDB, Power BI (DAX), MS Excel, Dashboards, Plotly, Streamlit
Machine Learning: scikit-learn, LightGBM, SHAP, Isolation Forest, NLP (VADER)
Generative AI: LLMs (Google Gemini), RAG, LangGraph, ChromaDB, Prompt-Injection Checks
Web & Databases: React, FastAPI, Node.js, Express, REST APIs, JWT Auth, PostgreSQL, MongoDB
Tools & DevOps: Git, GitHub Actions (CI/CD), Docker, pytest

Certifications

- SQL and Relational Databases 101 – Cognitive Class (IBM Skills Network) **Oct 2026**
- Python (Basic) Certification – HackerRank **Aug 2026**